DEPARTMENT OF COMPUTER APPLICATIONS
PSG COLLEGE OF TECHNOLOGY
MCA G1 & G2, SEMESTER 1

26MX16 - PROGRAMMING AND DATA STRUCTURES LABORATORY PRACTICE SHEET

Electricity Bill Calculation System

1. Problem Statement
Develop a program to calculate the monthly electricity bill of a consumer based on the number of units consumed. The system should accept consumer details and units consumed, apply the billing rules and generate the final payable amount.

2. Functional Requirement
The system shall:
- Accept consumer details.
- Accept the number of units consumed.
- Calculate the charge based on the applicable slab.
- Add the fixed charge.
- Apply surcharge, if applicable.
- Display the final electricity bill.

3. Business Rules
The following tariff rules are applicable:

Units Consumed	Rate
0–100	₹1.50/unit
101–300	₹2.50/unit
301–500	₹4.00/unit
Above 500	₹6.00/unit

Additional rules:
- Fixed charge = ₹100
- If the bill amount exceeds ₹2,000, a 5% surcharge is applied.
- Negative unit consumption is invalid.

Business Rule: These are real-world policies that the electricity provider expects the software to follow.

4. Constraints
Students should consider:
- Units consumed ≥ 0
- Consumer ID should not be empty.
- Units should be within the acceptable range of the billing system.
- Monetary values should be displayed with two decimal places.

Constraint: A limitation that the input or solution must satisfy.

5. Assumptions
For this laboratory exercise, assume:
- Billing is calculated for one month.
- There is only one consumer per program execution.
- The given tariff rates remain unchanged during the calculation.
- No additional taxes or subsidies are considered.

Prepared by J Gowri Thangam (JGT)

JGT

Assumption: A condition accepted for the purpose of solving the problem when the actual requirement is not fully specified.

6. Input Process Output (IPO) Analysis

INPUT **What do we receive?**	**PROCESS** **What does the program do?**	**OUTPUT** **What do we display?**
Consumer ID	Read consumer details	Consumer ID
Consumer Name	Read units consumed	Consumer Name
Units consumed	Check whether units are valid	Units consumed
	Identify the applicable tariff slab	Charge for the tariff slab
	Calculate charge	Fixed charge
	Add fixed charge	Surcharge, if applicable
	Check surcharge condition	
	Calculate surcharge	
	Calculate total bill	Final EB Bill

7. Input Specification

The program should accept:

- Consumer ID
- Consumer Name
- Units Consumed

Example:
Consumer ID: C1025
Consumer Name: Arun
Units Consumed: 350

8. Output Specification

The program should display:

```
Electricity Bill
------------------------
Consumer ID      : C1025
Consumer Name    : Arun
Units Consumed   : 350
Charge           : ₹...
Fixed Charge     : ₹100
Surcharge        : ₹...
Total Bill       : ₹...
```

9. Algorithm / Flowchart / Solution Design

Students should prepare the algorithm before coding:

START
↓
Read consumer details
↓
Validate units

↓
Determine applicable tariff
↓
Calculate charge for applicable tariff
↓
Add fixed charge
↓
Check surcharge condition
↓
Calculate total bill
↓
Display electricity bill
↓
STOP

10. Test Cases (Global / Local / Edge Cases)

Students must consider:

Case	Input	Expected Handling
Minimum	0 units	Valid/zero charge
Boundary	100 units	First slab
Transition	101 units	Next slab
Boundary	300 units	Second slab
Transition	301 units	Third slab
Boundary	500 units	Third slab
Transition	501 units	Fourth slab
Invalid	-10 units	Reject input
Surcharge boundary	Bill = ₹2,000	No surcharge
Above threshold	Bill > ₹2,000	Apply 5% surcharge

Edge Case: A special, extreme, or boundary situation that may expose errors in the program.

Sample Test Case Specification

Students must prepare at least 5 test cases.

TC ID	Units	Purpose	Expected Result
TC01	50	Normal case	First slab
TC02	100	Boundary case	First slab
TC03	101	Boundary transition	Second slab
TC04	500	Boundary case	Third slab
TC05	-10	Invalid case	Error message

11. Defect / Error Log

Students record errors encountered during implementation.

Error ID	Error/Defect	Type	Root Cause	Corrective Action
E01	Wrong slab selected	Logical	Incorrect condition	Corrected if-else order
E02	Negative units accepted	Validation	Missing input check	Added validation

12. Post-Implementation Review (PIR)

PIR Category	PIR Question	Response
Requirement	Does your program satisfy all the specified requirements, business rules, constraints, and assumptions? Identify any gap	
Testing	Did all the test cases validate functionality and correctness of your solution? Identify any missing test case or boundary condition.	
Defect	What error or defects did you encounter during implementation and testing? Analyses their probable root cause and recommend appropriate corrective measures.	
Improvement	How effectively does the software meet user needs in terms of usability, reliability, and error handling? Suggest one improvement	

JGT

DEPARTMENT OF COMPUTER APPLICATIONS
PSG COLLEGE OF TECHNOLOGY
MCA G1 & G2, SEMESTER 1

EXERCISES:

1. Write a C program to find all fractions with two-digit numerator and denominator for which "Sleepy's technique work correctly.
   Example.
   Sleepy's Technique
   26 2
   ----- = -----
   65 5
2. Consider an array of numbers from 1 to N. In this array, one of the number gets duplicated and one is missing. Write a C program to find out both.

HOMEWORK EXERCISE:

1. ATM Withdrawal System
2. IT Helpdesk Ticket Priority System
3. Online Shopping Bill Calculator

Self-Assessment:

At the end of each laboratory session, assess your learning and performance by completing the self-assessment table given below:

Reflection Question	Student Response
What did I learn today?	
What was the most challenging part?	
How did I overcome the challenge?	
What would I do differently next time?	
Which concept do I need to practice more?	

JGT